Lab 1: Intro to Networks, Unix and Python Primer

Points: 100 points – 5 points for progress check in lab

Due: Saturday 10/5/2024 at 11:59pm

Deliverables and Submission: 2 files

Naming convention: precede filenames with **your CruzID** (before '@' in your email: id@ucsc.edu):

1. <YourCruzID>-lab1.pdf
   Submit on Gradescope: a PDF file with each problem number and statement followed by your corresponding answer to each question.
2. <YourCruzID>.py
   Submit on Canvas: Python script in Question 9

GRADESCOPE POLICY reminder: Submissions on Gradescope will not be graded if the problem regions are not marked - see your TA or tutor if you need help with this.

Late Policy: Be aware of Late Policy (see Syllabus)

Required Screenshots:

Q1: a,b,c	Q4	Q9 Python Script run
Q2: b,c	Q5a	
Q3	Q6a	

Resources for this lab:

Linux: http://linux-training.be/index.php?nav=fundamentals (Links to an external site.)

Python: https://python.land/python-tutorial

UCSC TimeShare : https://its.ucsc.edu/unix-timeshare/tutorials/how-to-connect.html

Traceroute, IP Addresses and basic networks:

- Computer Networking: A Top-down Approach
  Chapter 1: introduction
  Chapter 1.4.3 : End-to-End Delay – Traceroute

Lab Environment Note: For this lab, all questions can be done using your own PC or the UCSC TimeShare

[20 points] Unix Practice

In this section you will learn or review some useful Unix commands. Test your commands and your Python program on a Linux machine. Here are instructions on using the UCSC Unix Timeshare for both MAC and Windows laptops:

https://its.ucsc.edu/unix-timeshare/tutorials/how-to-connect.html

1. [6 pts] Practice with directories and files
   a. From a shell/ command prompt on either a linux device or a PC, what is the complete command to log in to the UCSC Unix Timeshare (unix.ucsc.edu)? Test your command by logging in and include a **screenshot**.

   b. What command can be used to list all the files and directories (including hidden files) in the present working directory? Run the command in one of your directories and include a **screenshot** of the command and the resulting output. (If you don't have files, create a few random ones, you will need them later on).

   c. What command can be used to list all the files and directories in the present working directory in chronological order (based on the last modified date)? Run the command and Include a **screenshot** showing the resulting output.

   d. What command is used to display the current directory?

   e. What command is used to create a directory named "prelab1" in **your** home directory? What is the complete path name to this directory?

   f. What command is used to move to the root directory?

2. [6 pts] Working with Files: Copy the two files file1.txt and file2.txt into your directory and use them for questions (a),(b) and (c).

   a. What is the single line command to display the contents of the two files?

   b. What complete command allows you to see the line by line differences between these two files? Run the command and include a **screenshot** showing the results of running the command.

   c. What complete command is used to concatenate these two files and store the output to a new file named file3.txt? Run the command and Include **screenshot** showing the command and the resulting contents of file3.txt

   d. What complete command would you use to move file3.txt from your current directory to a directory having relative path "../dir2"?

   e. What complete command would you use to delete a file named file1.txt? Can the file be recovered once this command is issued? Why or why not? **Explain.**

   f. What command would you use to search for the string "vote" in "file3.txt" in the same directory as "file3.txt" is located in?

3. [8 pts] Moving files around the network:
   Create a directory named “network_log” (as shown below) in your home directory on the Unix Timeshare Server. In that directory, create the structure below with three .txt files, a png and a pdf file.

```
<user_directory>
└── network_log
    ├── f1.txt
    ├── f2.txt
    ├── f3.txt
    ├── a.png
    └── b.pdf
```

   What is the complete command to non-recursively copy all of the .txt files into your local computer’s current working directory? (This can either be on your own laptop or the lab computer.)

   Using this command, demonstrate that your command works by displaying a **screenshot** of the directory listing on the Unix server **and** then after the copy, show the directory listing on your local machine.

[35 points] Traceroute

The Traceroute program sends a series of probes (messages carried in a packet) from a source to a destination over the network. Each probe travels part-way to the destination until finally the destination is reached. The probes travel “hop by hop”, meaning from one router to the next on the path to the destination. In each iteration the probes incrementally get one hop closer to the destination until finally, in the last iteration, the destination is reached.

Note: When working on the problems, if Traceroute is hanging and doesn’t complete, talk to your TA. Multiple lines with *** and no completion is **not** what we are looking for in this exercise and will not receive credit. *** most likely means the router receiving the probe is not responding. Overall the majority of lines must display a normal response from the router.

4. [5 pts] Basic Traceroute
   Run Traceroute between your computer and www.vote.org. Include a **screenshot** of the output and answer the following questions based on the output in your screenshot:

   a) What does each of the numbered lines represent?

   b) Choose one of the output rows and circle the 3 time measurements displayed. Explain what they represent.

   c) How do you expect the time measurements in row #1 to differ from row #2? Greater? Less? The same? Explain your reasoning by focusing on the operation of each row in the program output.

   d) Define hop count. Based on your definition, how many hops are there between your computer and the final destination? Mark up your screenshot as needed.

5. [10 pts] North America: Run Traceroute between your computer and a destination on this continent (North America) at three different run times: morning, mid-day and night. Make sure that you are in the SAME location - i.e. at home or at school etc. for each of the runs. Record your results in the table below capturing the time to reach the final destination.

a) Include a **screenshot** of Run #1 (only) and circle all data used to calculate the average RTT to the destination host.

Destination Hostname: apple.com					**Location**: A tool such as IP Location Finder can be used to determine the approximate geographic location of a host.	
		Time to reach final destination				
Run #	Execution Time	RTT 1	RTT 2	RTT 3	Average RTT to final destination	# hops
1 (morning)	0.221	13.965	5.213	6.972	8.717	8
2 (mid-day)	5.328	4.897	11.650	8.673	8.407	8
3 (night)	10.354	11.575	11.041	12.173	11.596	8

b) Did the number of routers in the path change during your 3 different runs? If so, why do you think it changed? If not, why do you think it didn't?

6. [5 pts] Different Continent: Repeat process to a destination on a different continent.

Destination Hostname: 117.52.4.46						
Location of host: Seoul, Korea						
		Time to reach final destination				
Run #	Execution Time	RTT 1	RTT 2	RTT 3	Average RTT to final destination	# hops
1 (morning)	42.476	152.256	145.441	144.201	147.299	14
2 (mid-day)	41.021	152.015	152.266	148.115	150.799	14
3 (night)	36.137	137.376	146.979	148.465	144.273	15

a) Include a **screenshot** of Run #1 (only) and circle all data used to calculate the average RTT to the destination host.

b) Try to identify an overseas link with increased delay and mark/label it in your screenshot.

7. [10 pts] General questions about your experiment:

a. Compare the average intra-continent and inter-continent results. Are your results what you would expect? Explain.

b. For a given source and destination, find a row in your table for which the reported RTT values are different. What could account for the difference in delay?

8. [5 pts] Draw the North America network path: Think about our "Model Network on Paper" from the lecture and consider the traceroute output in Question 5. Make a drawing of the network. Make sure your drawing uses Nodes and Edges (routers and links) and include your computer as the starting Terminal.

- Identify your Access Network – mark and label it on your drawing. (An Access Network is the network used to connect to the Internet.)
- Routers with similar names and/or similar IP addresses can be considered as part of the same ISP. Put a dotted circle around any such groupings of nodes in your drawing to indicate these networks.
- Try to identify the number of ISPs that your packets traversed from the source to destination and write this number on the drawing.

[40 pts] Practice with Python

References: Refer to the Python tutorial above and to the previous section for Traceroute.

Need Python help?: The small group tutors are hosting Python tutorials all week!

9. Every time you surf the web, your browser displays a web document that is hosted on a web server. In order to display the web document, a request message must first be sent from your computer to the web server. Then, when a response message is returned and received by the browser, the requested page is displayed. Sometimes the messages must be forwarded by many routers on the path from your computer to the web server. In this exercise we will use **traceroute** to look at the "length" of such paths.

Write a simple utility program that accepts a web server name as input and then invokes the **traceroute** program to determine the number of hops from the source (where the program is invoked) to the web server.

Program operation:

- prompt the user for a host name (a web server)
- Hard coding is not allowed:
    - read the host name from the command line
    - print the output to **stdout**

- invoke the 'traceroute' command to determine the number of hops to a web server. Allow Traceroute to run for 20 seconds before checking its output.

Program Requirements:

- Use Python3

- Your program must run on the **UCSC Unix Timeshare** (see above). If your program does not run on this system we cannot grade it.

- Your program must include at least one function

- Error Checking: Does Traceroute terminate? If traceroute is unable to terminate, print an error message that states "Traceroute: failed". Recommendation: Before running your script, verify the expected Traceroute for the destination web server.

- Command line input and displayed output should be as indicated in these examples:

	Command line:	Expected Output
Example 1	Enter a server name: www.abcnews.com	X hops to <webserver name>, e.g. 8 hops to www.abcnews.com
Example 2	Enter a server name: <unreachable host>	Traceroute: failed
Example 3 [5 points extra credit]	Enter a server name: <hostname> (timeout occurs)	Traceroute: failed

Capture the following in a **screenshot**: Run traceroute from the command line. Then run your script and show the output line in your screenshot. Briefly compare and discuss the results.

Don't forget to upload your program to Canvas!